EXACT FUNCTORS ON PERFECT COMPLEXES OF REGULAR PROPER SCHEMES HAVE ADJOINTS

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Editorial note. Existence delegates to the BVdB representability theorem (statement in chapter carries an external citation, uncardable here); the hypothesis verification is in-span; the left adjoint is mirror-level.

Allowed standard prerequisites (editorial).

- BVdB representability. theorem-bondal-van-den-bergh (same chapter statement carries an external citation, uncardable here). Representability of cohomological functors with finite total dimensions.
- Finiteness input. lemma-finiteness (same chapter, judged citation-only, not carded). The finite total dimension hypothesis.

Source reference. Original reference 0A8D: focused-seeds/geometry/S9-eqv-always-right-adjoints/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYC: sources/stacks-equiv/pinned-original-source.tex, source lines 838–838. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0H4A: sources/stacks-equiv/pinned-original-source.tex, source lines 1062–1062. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 0FYH: sources/stacks-equiv/pinned-original-source.tex, source lines 1032–1032. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X and Y be proper schemes over k. If X is regular, then any k-linear exact functor $F : D_{perf}(\mathcal{O}_X) \to D_{perf}(\mathcal{O}_Y)$ has an exact right adjoint and an exact left adjoint.*

Proof. If an adjoint exists it is an exact functor by the very general Derived Categories, Lemma 0A8D.

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let us prove the existence of a right adjoint. To see existence, it suffices to show that for $M \in D_{perf}(\mathcal{O}_Y)$ the contravariant functor $K \mapsto \mathrm{Hom}_Y(F(K), M)$ is representable. This functor is contravariant, k-linear, and cohomological. Hence by Theorem 0FYH it suffices to show that

$$\sum\nolimits_{i \in \mathbf{Z}} \dim_k \mathrm{Ext}^i_Y(F(K), M) < \infty$$

This follows from Lemma 0FYC.

For the existence of the left adjoint we argue in the same manner using Lemma 0H4A in stead of Theorem 0FYH. $\square$